Design of the Low Vision Quality-of-Life Questionnaire (LVQOL) and Measuring the Outcome of Low-vision Rehabilitation

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• PURPOSE: To design and validate a vision-specific quality-of-life assessment tool to be used in a clinical setting to evaluate low-vision rehabilitation strategy and management.

• METHODS: Previous vision-related questionnaires were assessed by low-vision rehabilitation professionals and patients for relevance and coverage. The 74 items selected were pretested to ensure correct interpretation. One hundred and fifty patients with low vision completed the chosen questions on four occasions to allow the selection of the most appropriate items. The vision-specific quality of life of patients with low vision was compared with that of 70 age-matched and gender-matched patients with normal vision and before and after low-vision rehabilitation in 278 patients.

• RESULTS: Items that were unreliable, internally inconsistent, redundant, or not relevant were excluded, resulting in the 25-item Low Vision Quality-of-Life Questionnaire (LVQOL). Completion of the LVQOL results in a summed score between 0 (a low quality of life) and 125 (a high quality of life). The LVQOL has a high internal consistency ($\alpha = 0.88$) and good reliability (0.72). The average LVQOL score for a population with low vision (60.9 ± 25.1) was significantly lower than the average score of those with normal vision (100.3 ± 20.8). Rehabilitation improved the LVQOL score of those with low vision by an average of 6.8 ± 15.6 (17%).

• CONCLUSIONS: The LVQOL was shown to be an internally consistent, reliable, and fast method for measuring the vision-specific quality of life of the visually impaired in a clinical setting. It is able to quantify the quality of life of those with low vision and is useful in determining the effects of low-vision rehabilitation. (Am J Ophthalmol 2000;130:793–802. © 2000 by Elsevier Science Inc. All rights reserved.)

THERE IS A NEED FOR A MEASURE OF THE OUTCOME OF low-vision rehabilitation to be able to argue the case of low-vision assessment within managed care plans, to improve services offered, and to secure and enhance funding for low-vision services.[1] There are significant individual, public health, and community consequences of reduced vision, such as increased cost of education, reduced personal income, and loss of productivity for those caring for or assisting the visually impaired. Relative to the costs of visual impairment, it has been suggested that the provision of low-vision rehabilitation services may be quite low. The prevalence of low vision is likely to increase significantly over the next decade because the percentage of elderly people, who are most often afflicted with low vision, is increasing.[1]

There are many measures taken in a routine low-vision assessment on which successful outcome has been rated. These measures include distance acuity,[2] near acuity,[3,4] reading speed and duration,[5,6] and comprehension and fluency.[5] The term "success" is often used in visual rehabilitation, based on whether a patient finds an aid beneficial (as defined by the patient) and uses it to solve one or more visual problems,[7,8] or failed to attend follow-up appointments.[9] Some studies have examined the ability of patients with low vision to perform simple tasks, such as identifying paper currency and clock reading, before and after low-vision rehabilitation, but the results vary with the examiner performing the assessment, and implementation is time consuming and expensive.[10]

As a more detailed answer to the question of how to measure outcome, subjective quality-of-life questionnaires have been developed. They aim to identify and quantify the impact of visual impairment on aspects of daily living, such as self-care, mobility, and dependency.[11] It is recognized that patients are not interested in improvements in a biomedical indicator, but rather in how treatment affects their quality of life.[12,13]

Accepted for publication June 2, 2000.

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This projected was funded by a grant from the Victorian College of Optometry, Victoria, Australia, and Vision Australia Foundation, Australia.

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0002-9394/00/$20.00
PII S0002-9394(00)00610-3

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TABLE 1. Sources of Items Used in the LVQOL

Questionnaire Authors	Condition	Grading Scales
Ross and associates (1984)[35]	Glaucoma	Ordinal 1–5
Mills and Drance (1986)[19]	Glaucoma	Yes, uncertain, no
Nilsson and Nilsson (1986)[6]	ARMD	Yes, no
Lowe and Drasdo (1992)[15]	Retinitis pigmentosa	Yes, no; 2 to 4 categories; open ended
Mangione and associates (1992)[34]	Cataract (ADVS)	Five categories
Leat and associates (1994)[16]	Low vision	Yes, no; 3 to 5 categories; open ended
Rubin and associates (1994)[36]	Old age visual impairment	Three categories
Steinberg and associates (1994a,b)[22,23]	Cataract (VF-14)	Yes (4 categories), no/not applicable
Ellwein and associates (1995)[13]	Blindness	Four categories/does someone help you
Parrish (1996)[12]	Glaucoma	Three to 5 categories; ordinal 0–10
Abrahamsson and associates (1996)[11]	Cataracts	Yes, no
Wu and associates (1996)[21]	Cytomegalovirus	Yes, no
Cleary and associates (1997)[20]	Optic neuritis	Five categories/not applicable
Javitt and associates (1997)[37]	Cataract	Ordinal 0–4 or 0–10; 5 categories; open ended
Lundstrom and associates (1997)[17]	Cataract (Catquest)	Yes, no; yes (3 categories), no; 2 to 5 categories/cannot say
Carta and associates (1998)[25]	ARMD, branch retinal vein occlusion, cataract, primary open-angle glaucoma	Ordinal 1–3 with descriptors

ARMD = age-related macular degeneration. ADVS = Activities of Daily Vision Scale.

Various simple questionnaires have been used to assess quality of life in low vision.[6,14–16] However, most have been developed using a population with reduced vision from potentially treatable conditions[11,17,18] or for people with a single ocular condition causing low vision,[12,15,19–21] so few are valid for the measurement of the outcome of low-vision rehabilitation in a general population with low vision. Visually related quality-of-life questionnaires, such as the VF-14 developed for cataract assessment,[22,23] have been modified to be appropriate for particular populations, but the validity of these modifications have not been examined. Measurement of quality of life with many of the questionnaires used in research studies is very time consuming, and therefore their clinical application would place a significant burden on patients.[24,25]

Hoppe and associates[26] examined the quality of life of patients with low vision before and after a low-vision assessment, finding no significant improvement had been achieved. However, the questionnaire used was not specific to patients with nontreatable visual loss and tapped traits not relevant to rehabilitation of the visually impaired. Scott and associates[27] found that quality of life in 156 patients was unchanged by low-vision rehabilitation when measured by a general health-related questionnaire (SF-36), but improved with vision-specific questionnaires. However, they noted that neither of the vision-specific questionnaires used (VF-14 and NEI-VFQ-51) had been designed specifically for a population with low vision.

The current study aimed to develop a questionnaire, using patients with a wide variety of conditions causing nontreatable vision loss, which addresses issues relating to quality-of-life reduction in those with low vision and is able to quantify the benefit of low-vision rehabilitation. The tool developed will be of great benefit to low-vision specialists if it enables fast and accurate audit of the outcome of low-vision rehabilitation and objectively examines of the benefits of different models of rehabilitation and care of the visually impaired.

METHODS

THE LITERATURE WAS REVIEWED TO COLLECT ALL THE subjective questionnaires previously designed for assessing quality of life (Table 1). The resulting questions were merged, so no items were repeated. The items were then assessed by a multidisciplinary team of low-vision rehabilitation staff (including ophthalmologists, optometrists, orthoptists, occupational therapists, welfare officers, audiologists, and those with low vision) for relevance: questions not relevant to the majority of patients with low vision were excluded; coverage: whether all areas of life affected by low vision, such as activities of daily living, social and emotional factors, were examined; face validity: whether questions appear on the surface to be measuring what they actually are.

Closed-end questions, with a specific set of responses, were used, because they can be consistently scored. The study protocol was approved by the Institutional Review Board, and informed consent was obtained from subjects participating in the experiment.

The resulting questionnaire had 74 questions that were

pretested to ensure correct interpretation by the users. The questionnaire was printed in writing of sufficient size (equivalent to N20) for most visually impaired people to read. Many different response options have been used by previous vision-related quality-of-life questionnaires (Table 1). An ordinate rather than a line scale was used for the questionnaire to allow implementation over the phone as well as in person and by post. A six-point scale was instituted to allow a reasonable level of discrimination, while not causing confusion among the mostly elderly population examined. Patients with low vision responded to items grading them between 5 (having no difficulty with the item because of their vision) and 1 (having a great difficulty with the item because of their vision) or as 0 (item could no longer be performed because of their vision). They also had the option to rate the question as "not relevant" if it was not relevant to them in their daily lives (given an average score so that those individuals who scored more items irrelevant did not have a lower summed questionnaire score and therefore an apparently worse quality of life). Missed questions (contained in less than 0.5% of questionnaires) were also given an average score.

The 74-item questionnaire was trialed on 150 patients with low vision randomly selected from the database of the Vision Australia Foundation. They had a wide range of common conditions causing low vision in a developed world country. Distance visual acuities ranged from 6/7.5 to 1/60 with visual field defects affecting the central 20 degrees of visual field in 53%. The questionnaire was completed on four occasions, each separated by 4 weeks (± 4 days) by the same implementation method each time (by post, phone, or in-person interviews). The aim of the initial trial of the questionnaire was to identify those items that were most reliable, internally consistent, not redundant, relevant to most subjects, and that the full range of response options were used. The 25 items that met these criteria formed the Low Vision Quality-of-Life Questionnaire (LVQOL) (Figure 1).

To assess the LVQOL ability to differentiate the visually impaired from those with normal vision, the LVQOL was completed by an age-matched and gender-matched control group (of 70 subjects) attending a large public optometric clinic in Melbourne for a routine eye examination.

The LVQOL was sent to 515 consecutive first-time patients of a low-vision clinic (which provides free rehabilitation services) before their appointment for rehabilitation. A second LVQOL was sent to patients after rehabilitation, which consisted of:

1. A 15-minute to 30-minute interview with a case manager to identify visual needs. Details on the patient's age, gender, living status, and primary condition causing visual impairment (as diagnosed by the referring ophthalmologist) were recorded.
2. A 60-minute low-vision assessment with an optometrist, which included a review of the patient's ocular history and visual complaints, establishment of visual goals, retinoscopy and refraction at distance and near, measurement of visual acuity, contrast sensitivity, and central visual field. Advice was given (for example, on lighting and contrast), and magnification aids were prescribed if appropriate. For each patient, the following clinical information was collected (consulting rooms had an illuminance of 500 lux): progression of visual difficulties as stated by the patient; distance visual acuity (Bailey-Lovie distance logMAR chart back-lit to a luminance of 160 cd/m^2) scored as the number of letters correctly identified with effort; near visual acuity (Practical Near Acuity Chart at 25 cm, read with the best distance correction and a +4.00-diopter near addition)[28]; magnification aids; presence of a visual field loss or disruption in the central 20 degrees field of view, measured using an Amsler chart at 14 cm; peak contrast sensitivity (Melbourne Edge Test, luminance 47 cd/m^2 at 40 cm)[29]; assessment of the patient's psychological status based on a reduced version of the Nottingham Adjustment Scale[30,31]; services from members of a multidisciplinary team as appropriate to their needs. The team includes orthoptists, occupational therapists, orientation and mobility instructors, audiologists, and welfare officers who offer assessment and assistance either at the clinic or at the patient's residence.[31]

Internal consistency was examined using the item-total correlation and Cronbach's α, reliability using the intraclass correlation coefficient, and principal components using varimax rotation factor analysis.[32] The summed score of the LVQOL items was calculated for analysis, as has been previously recommended for quality-of-life analysis.[30] Analysis of variance was used to examine overall effects between groups and Tukey's pairwise multiple comparison test to assess individual group differences. Correlation models were examined using basic subsets.

RESULTS

• **SELECTION OF APPROPRIATE ITEMS FOR THE LVQOL:** To select the most appropriate items for the LVQOL from the initial pool of questions, the following exclusion criteria were implemented:

Frequency of Endorsement and Discrimination. The proportion of subjects who gave each response alternative was calculated. This allowed the exclusion of five questions that were not relevant to a significant proportion of the subjects (taken at 33%). They were related to using public transport alone, playing sports and games, going to the cinema or theater, and how much benefit was gained by talking to a peer worker. Items with very low or high endorsement were also eliminated, because little is learned

Distance Vision, Mobility and Lighting

How much of a problem do you have:	GRADING None		Moderate		Great		
With your vision in general	5	4	3	2	1	x	n/a
With your eyes getting tired (e.g only being able to do a task for a short period of time)	5	4	3	2	1	x	n/a
With your vision at night inside the house	5	4	3	2	1	x	n/a
Getting the right amount of light to be able to see	5	4	3	2	1	x	n/a
With glare (e.g dazzled by car lights or the sun)	5	4	3	2	1	x	n/a
Seeing street signs	5	4	3	2	1	x	n/a
Seeing the television (appreciating the pictures)	5	4	3	2	1	x	n/a
Seeing moving objects (e.g. cars on the road)	5	4	3	2	1	x	n/a
With judging the depth or distance of items (e.g. reaching for a glass)	5	4	3	2	1	x	n/a
Seeing steps or curbs	5	4	3	2	1	x	n/a
Getting around outdoors (e.g. on uneven pavements) because of your vision	5	4	3	2	1	x	n/a
Crossing a road with traffic because of your vision	5	4	3	2	1	x	n/a

Adjustment

Because of your vision, are you:	No		Moderately		Greatly		
Unhappy at your situation in life	5	4	3	2	1	x	n/a
Frustrated at not being able to do certain tasks	5	4	3	2	1	x	n/a
Restricted in visiting friends or family	5	4	3	2	1	x	n/a

	Well				Poorly	Not explained
How well has your eye condition been explained to you	5	4	3	2	1	x

Reading and Fine Work

With your reading aids / glasses, if used, how much of a problem do you have:	None		Moderate		Great		
Reading large print (e.g. newspaper headlines)	5	4	3	2	1	x	n/a
Reading newspaper text and books	5	4	3	2	1	x	n/a
Reading labels (e.g. on medicine bottles)	5	4	3	2	1	x	n/a
Reading your letters and mail	5	4	3	2	1	x	n/a
Having problems using tools (e.g. threading a needle or cutting)	5	4	3	2	1	x	n/a

Activities of Daily Living

With your reading aids / glasses, if used, how much of a problem do you have:	None		Moderate		Great		
Finding out the time for yourself	5	4	3	2	1	x	n/a
Writing (e.g. cheques or cards)	5	4	3	2	1	x	n/a
Reading your own hand writing	5	4	3	2	1	x	n/a
With your every day activities (e.g. house-hold chores)	5	4	3	2	1	x	n/a

FIGURE 1. The LVQOL. Patients are asked to complete the questions by circling the number most appropriate to how they feel. If they can no longer perform the task because of their vision, they are to circle "x," and if they do not perform the task for nonvisual reasons, to circle "n/a".

by knowing how a person actually responded if one can predict the answer to a high accuracy. Such questions do not improve a scale's psychometric properties and may detract from them while making the test longer. Therefore, items that were endorsed by over 65% of the subjects as of little difficulty or of great difficulty were excluded. Some of the issues that the majority of patients with low vision reported having little difficulty with were perceiving colors, eye pain or discomfort, performing household chores, and making or eating a meal. An issue for which the majority of patients with low vision had significant difficulty was reading television subtitles, and this item was excluded because it did not assist in differentiating between subjects. The discrimination ability index was also calculated to identify if a person who has a high rating on a sub-scale is more likely to endorse a particular item of that sub-scale.[32] However, this measure did not further discriminate between questionnaire items.

Homogeneity and Internal Consistency. A scale must be homogenous, tapping different aspects of the same attribute and not different parts of different traits. Therefore, items should be moderately correlated to one another, and each should correlate with the total scale score. The most sensible index of a particular sub-scale would appear to be a simple summing of the scores.[32] Such a linear model is valid only if all items are measuring the same trait. If items are highly correlated, latter questions add little information, suggesting redundancy with a possible loss of content validity. If the correlations are too low, then the questions are tapping different traits. The two measures calculated were the item-total correlation (correlation of an individual item with its sub-scale without that item) and Cronbach's α (a measure of internal consistency that should be between 0.70 and 0.90).[33] Item-total correlation was less than 0.20 for four questions regarding general health, the impact of the low-vision assessment, and the feeling of independence. These questions were therefore excluded from the questionnaire, because they do not tap the same trait as the rest of the scale. Cronbach's α resulted in the removal of no further questions.

Reliability. Reliability is fundamentally a way to reflect the amount of error, both random and systematic, inherent in any measurement. The test-retest reliability was calculated using the results from the four completions of the questionnaire. There appears to be no consensus over what interval the test-retest reliability should be conducted, although it should be sufficiently short to be able to assume the underlying process is unlikely to have changed and sufficiently long that patients do not remember their previous ratings. The progression of visual impairment will vary greatly between patients in this population, depending on the condition causing low vision. As low-vision rehabilitation is a process that can take several sessions to achieve, a 4-week interval was used to select those items that remained stable with time. Measures of mild and presumably variable pathological states typically have coefficients of approximately 0.7. Using a removal criteria of reliability less than 0.60, a final 20 questions were removed from the questionnaire. They examined such issues as use of a telephone, purchasing an item in a store, adapting to changing light levels, feelings of embarrassment, seeing faces, and needing assistance to perform a task.

Factor Analysis. A vision-related quality-of-life questionnaire can either test different aspects of vision using sub-scales, such as the Activities of Daily Vision Scale,[34] or produce a summed score of the patients' quality of life, such as the VF-14.[22,23] The LVQOL is of the latter type, with the 25 items producing a summed score between 0 and 125 (the higher the score, the higher the quality of life). Analysis of the results by implementation method did not alter the final LVQOL items. The LVQOL summed score showed a high internal consistency (Cronbach's α = 0.88) and a good reliability (0.72). Factor analysis using varimax rotation identified up to four principal factors in the LVQOL, each with a high internal consistency (Cronbach's α greater than 0.80). The LVQOL items were grouped into these sub-sections (general vision, mobility, and lighting issues; psychological adjustment; reading and fine work; activities of daily living).

• **ASSESSING THE VISION SPECIFIC QUALITY OF LIFE OF SUBJECTS WITH NORMAL VISION:** The LVQOL was designed to measure the quality of life of those with visual impairment and must therefore differentiate those with low vision from those with normal vision. The LVQOL was completed by an age-matched (68.3 $\pm$ 10.5 years) and gender-matched (39% male) control group. The average ($\pm$ SD) LVQOL score for those with normal vision was significantly higher than those from the population with low vision (100.3 $\pm$ 20.8 versus 60.9 $\pm$ 25.1; $P < .001$). Over 67% of those with normal vision had a LVQOL score of 95 or greater compared with less than 20% of those with low vision (Figure 2).

• **QUANTIFYING THE BENEFIT OF LOW VISION REHABILITATION:** The LVQOL was then used to determine whether improvement in quality of life occurs after low-vision rehabilitation. Of the 515 referrals to the low-vision clinic who received the LVQOL with their appointment letter, 65% returned the completed LVQOL at their appointment. A postrehabilitation questionnaire was sent to those who had returned the prerehabilitation questionnaire (accompanied by a stamped addressed reply envelope), and this was returned by 83%. There was no significant difference in visual function or rehabilitation received by each of these groups (Table 2). There was a significant improvement ($P < .001$) in LVQOL score of 6.8 $\pm$ 15.6 (a 17% improvement compared with the LVQOL score of those with normal vision). The reading

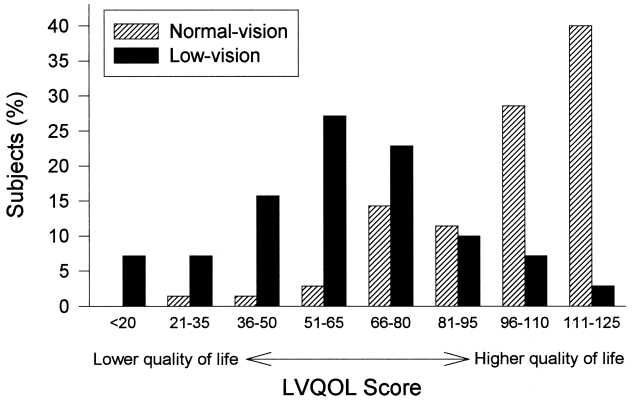


FIGURE 2. The distribution of LVQOL scores for age-matched and gender-matched groups of patients with normal and low vision (n = 70).

and fine work subscale was most improved (average, 0.54 ± 0.07 of 5), but the other subscales (general vision, mobility, and lighting 0.17 ± 0.05; psychological adjustment 0.21 ± 0.07; activities of daily living 0.24 ± 0.07) also showed significant improvement with low-vision rehabilitation ($P < .001$). Low Vision Quality-of-Life Questionnaire summed scores covered the entire range of possible values and were normally distributed (Figure 3).

The LVQOL score before or after rehabilitation was found to be correlated to distance and near acuity (with the present prescription or best corrected) and contrast sensitivity, but not the patient's age (Table 3). A basic subset model showed that distance best-corrected visual acuity and contrast sensitivity were the principal functional vision contributors, accounting for 32% of the variance in the prerehabilitation LVQOL measures and 25% of the postrehabilitation LVQOL measures. The improvement in LVQOL score was not correlated with acuity measures, contrast sensitivity, or a patient's age. The LVQOL score and improvement with rehabilitation was not related to the patients' gender, the condition causing low vision, how they felt their condition had progressed, or whether they already used low-vision aids before rehabilitation (Table 4). Those with a visual field loss affecting the central 20 degrees had a lower before and after rehabilitation quality of life than those who did not have a visual field loss. A magnifier alone was more commonly prescribed to those with a lower quality of life. Those making use of the multidisciplinary low-vision rehabilitation team had a lower before and after rehabilitation quality of life than those who did not but also had a significantly larger improvement in quality-of-life score with rehabilitation. Those classified as not having accepted their visual loss tended to have a lower quality of life and less improvement with rehabilitation.

DISCUSSION

THERE IS A NEED FOR A QUALITY-OF-LIFE ASSESSMENT TOOL for clinical low-vision practice to gain a better understanding of patients' needs and goals and whether they have been met by rehabilitation. Excluding items that had a low or high endorsement, a poor internal consistency, and that were not reliable allowed the pool of 74 items, selected by a multidisciplinary team of rehabilitation professionals and patients with low vision, to be reduced to a 25-item LVQOL. Implementation of the LVQOL produces summed score between 0 and 125 (the higher the score, the higher the quality of life) with a high internal consistency (Cronbach's $\alpha = 0.88$) and a good reliability over a 12-week period (0.72). The use of a single index score has been recommended for quality-of-life assessment, but it loses information about specific variables.[30] Splitting the LVQOL into its four principal sections will reduce the amount of information lost, and with only 25 questions, the individual item responses can easily be recorded for reference regarding specific variables if needed.

A single questionnaire is unlikely ever to cover the whole range of quality-of-life issues in visual impairment for every individual, because the extreme length of a totally comprehensive questionnaire would preclude its use in clinical and many research settings.[18] However, the LVQOL provides useful clinical information that, in addition to functional measures of vision, such as visual acuity, contrast sensitivity, and visual field, describe the visual status of an individual. Such information can be used to assess the outcome of low-vision rehabilitation and will help to identify those cases where further training, service provision, or re-assessment may be needed to improve the quality of life of an individual.

The LVQOL was shown to differentiate a population with low vision from an age-matched and gender-matched group attending an optometric clinic. Some of the age-matched and gender-matched group will have had early cataract, blepharitis, dry eye, and other diseases affecting vision. Because they were all on state benefit, they were also likely to be of a lower socio-economic status than the clinic population with low vision. Therefore, it is not surprising that some members of this group rated their vision-related quality of life as reduced, but on the whole their quality of life was much higher (approximately 40 units of the 125 range of the LVQOL) than the population with low vision. Therefore, the 6.7-point average improvement in LVQOL score equates to a 17% improvement in quality of life. This equates well with the 5.4-point improvement in score (on a 100-point scale) measured by the VF-14 and would seem more sensitive than the 22.2-point improvement in overall score (on a 1300-point scale) measured by the NEI-VFQ with low-vision rehabilitation.[27] However, in that study postrehabilitation measures were taken at 3 months compared with 1 month with the LVQOL, and with an average of 3.4 magnifiers

TABLE 2. Demographic and Visual Function Characteristics of Population With Low Vision Receiving Rehabilitation

	Failed to Return Before Rehabilitation LVQOL	Returned Before but not After Rehabilitation LVQOL	Returned Before and After Rehabilitation LVQOL
Number of patients	181	56	278
Age (years)	74 ± 17	74 ± 18	78 ± 11
Gender (% males)	38	37	34
Living alone (%)	34	37	34
Progressive eye condition (%)	63	66	61
Visual need of reading or writing (%)	80	72	80
Best-corrected distance visual acuity (logMAR)	0.58 ± 0.35	0.56 ± 0.33	0.56 ± 0.34
Best-corrected near visual acuity (logMAR)	0.67 ± 0.33	0.69 ± 0.36	0.64 ± 0.32
Contrast sensitivity (dB)	11.6 ± 4.2	11.0 ± 3.9	11.1 ± 3.6
Field loss in central 20 degrees (%)	39	39	42
Magnification aids prescribed (%)	42	45	40
Used multidisciplinary rehabilitation team (%)	65	65	68
Psychological acceptance of vision loss (%)	68	75	76
Pre-rehabilitation LVQOL score	—	66.3 ± 24.5	63.5 ± 23.6
Averages ± SD	Total number of subjects = 515		

dB = decibels; LVQOL = Low Vision Quality of Life Questionnaire.

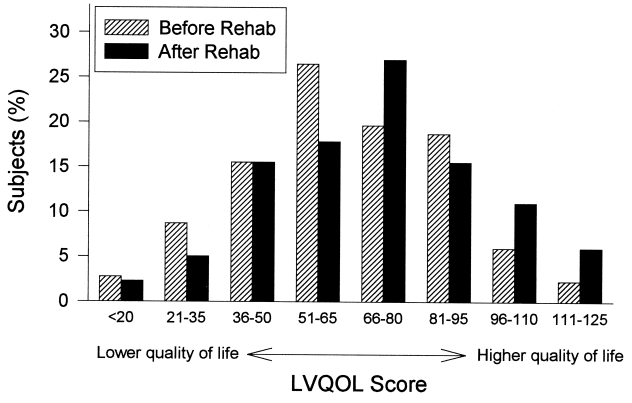


FIGURE 3. The distribution of LVQOL scores before and after low-vision rehabilitation (n = 334).

prescribed per person, the population and rehabilitation methods would appear to differ significantly from that of the Kooyong clinic.[31]

As expected, the LVQOL score was related to functional measures of vision, such as distance and near visual acuity, contrast sensitivity and field loss, but in combination these factors accounted for only 32% of the variance. Age was not related to the LVQOL score, which again confirms it is not measuring age-related visual or health-related factors (such as presbyopia). Improvements in quality of life with rehabilitation were not associated with patient demographics (age and gender), visual function (visual acuity and contrast sensitivity), or the disease causing low vision. Therefore, it would seem that people attending a low-vision clinic with visual loss from any condition receive significant benefit to their quality of life from low-vision rehabilitation. The age-matched and gender-matched group without low vision showed a quality-of-life improvement of just 0.8 ± 11.6 on the LVQOL after an eye examination.

Those patients judged to have reached a psychological state of realistic acceptance had a higher quality of life and a greater improvement with low-vision rehabilitation than those who were still adjusting to their visual loss, as would be expected. This finding emphasized the need for a basic assessment of a patient's psychological state to provide important additional information to functional visual measures. Those subjects that made use of the multidisciplinary rehabilitation team had a lower quality of life, but a greater improvement with rehabilitation, than those who did not make use of the team. This may have been because of patients' self-selection of whether they would accept the additional (free) services offered to them but would indicate that the multidisciplinary approach to low-vision rehabilitation is an important way to maximize quality of life, especially of those with greater functional loss. Further research is needed to assess what elements of rehabilitation, such as number of clinic visits, are necessary for improving the quality of life of the visually impaired.

By using quality-of-life measurement tools in a clinical and research setting, it should be possible to determine changes in the way in which low-vision rehabilitation can be performed to achieve greater improvements in a patient's quality of life. However, as long as the condition causing the reduced vision remains untreatable, a 100% improvement cannot be expected. Examining the link between quality of life and measures of cost benefit, such as a reduction in falls, may help to expand the provision of

TABLE 3. Correlation Between LVQOL Scores Before and After Low-vision Rehabilitation, Patient Demographics, and Functional Vision Measures

	Prerehabilitation LVQOL Score	Postrehabilitation LVQOL Score	LVQOL Improvement With Rehabilitation
Age (years)	0.04	0.01	0.04
Distance visual acuity with own prescription (logMAR)	0.47*	0.45*	−0.01
Near visual acuity with own prescription (logMAR)	0.47*	0.42*	0.02
Best-corrected distance visual acuity (logMAR)	0.50*	0.48*	−0.00
Best-corrected near visual acuity (logMAR)	0.44*	0.45*	−0.06
Contrast sensitivity (dB)	−0.43*	−0.40*	−0.02

Abbreviations as in Table 2.
*Indicates correlation significant to $P < .001$.
Total number of subjects = 278.

TABLE 4. Mean LVQOL Scores Before and After Low-vision Rehabilitation With Patient Demographics and Rehabilitation Characteristics

	Prerehabilitation LVQOL Score	Postrehabilitation LVQOL Score	Improvement with Rehabilitation
Gender			
Male	63.9 ± 23.0	70.4 ± 26.3	6.5 ± 16.6
Female	63.5 ± 24.0	70.6 ± 24.5	7.1 ± 15.0
Eye condition			
ARMD	65.1 ± 23.1	71.7 ± 22.8	6.6 ± 14.6
Glaucoma	61.1 ± 20.1	68.4 ± 23.3	7.3 ± 19.4
Diabetic retinopathy	59.6 ± 24.5	65.9 ± 29.3	6.3 ± 15.5
Other conditions	62.4 ± 25.0	69.4 ± 30.1	7.1 ± 15.8
Multiple conditions	60.8 ± 27.8	68.5 ± 28.0	7.7 ± 17.7
Eye condition progression			
None	65.8 ± 23.8	70.8 ± 26.3	5.0 ± 16.9
Steady	63.7 ± 22.9	70.4 ± 24.5	6.7 ± 15.1
Episodic	61.9 ± 23.4	71.1 ± 24.3	9.2 ± 17.4
Visual field			
No central 20-degree loss	70.4 ± 23.4	80.1 ± 21.0	9.7 ± 17.9
Central 20-degree loss	61.1 ± 24.9*	66.3 ± 28.7*	5.1 ± 15.4
Initial low-vision aids			
None used	65.3 ± 23.0	71.4 ± 25.9	6.1 ± 14.9
Uses low-vision aids	62.0 ± 24.1	69.3 ± 24.5	7.3 ± 16.1
Psychology			
Unrealistic	57.5 ± 23.4	61.6 ± 27.3	4.0 ± 18.1
Realistic acceptance	65.3 ± 23.2	73.0 ± 23.5*	7.7 ± 14.7
Prescribed low-vision aids			
None	62.7 ± 26.2	69.5 ± 28.0	6.8 ± 16.8
Magnifiers	59.9 ± 20.8*	67.1 ± 22.7	7.2 ± 15.7
New glasses	73.6 ± 19.5	77.8 ± 21.7	4.2 ± 15.5
Magnifiers + glasses	67.8 ± 29.9	76.2 ± 30.3	8.4 ± 11.3
Multidisciplinary team			
No need for team	70.2 ± 21.5	74.3 ± 24.8	4.1 ± 13.6
Used team	60.6 ± 23.9*	68.6 ± 25.2*	8.0 ± 16.4*

Abbreviations as in Tables 1 and 2.
Average LVQOL score ± SD.
Postrehabilitation refers to evaluation 6 weeks after initial rehabilitation at the low-vision clinic, regardless of how many visits were made to the low-vision clinic over that period.
*Indicates $P < .01$.

low-vision rehabilitation services, despite limited healthcare funds. The lack of difference in demographics and functional visual measures between those who failed to complete the LVQOL in a clinical setting and those completing both prerehabilitation and postrehabilitation measures suggests that postal implementation of a quality-of-life questionnaire for those with visual impairment does not bias the results collected. Translating the LVQOL into different languages should improve the response rate and make it easier to complete for patients with a first language other than English. To improve response rates further, those who attend the clinic without completing the questionnaire could be requested to fill out the LVQOL while waiting to be seen, and a telephone call 2 months after rehabilitation could increase the response rate among those who fail to complete the postrehabilitation questionnaire.

The LVQOL takes 5 to 10 minutes to complete and would therefore appear to not put undue strain on a person with low vision. The average LVQOL score (approximately 60 out of 125) demonstrates that a population of patients with low vision perceived their functional status and quality of life to be markedly impaired. Care was taken to exclude culture-specific aspects of lifestyle, such as technology, from the LVQOL to make it applicable in a wide variety of cultural settings, so it could be used worldwide without need for alteration of content.

In conclusion, the LVQOL has been shown to be an internally consistent, reliable, and sensitive measure of the quality of life in those with untreatable visual loss. It can be used in a clinical setting to provide important additional information to conventional functional visual measures, without significant burden on patients. It is able to quantify the quality of life of those with low vision and is useful in determining the effects of low-vision rehabilitation.

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